

Week 8
Control (Part 1)

ENN581 Robot Motion, Control and Planning

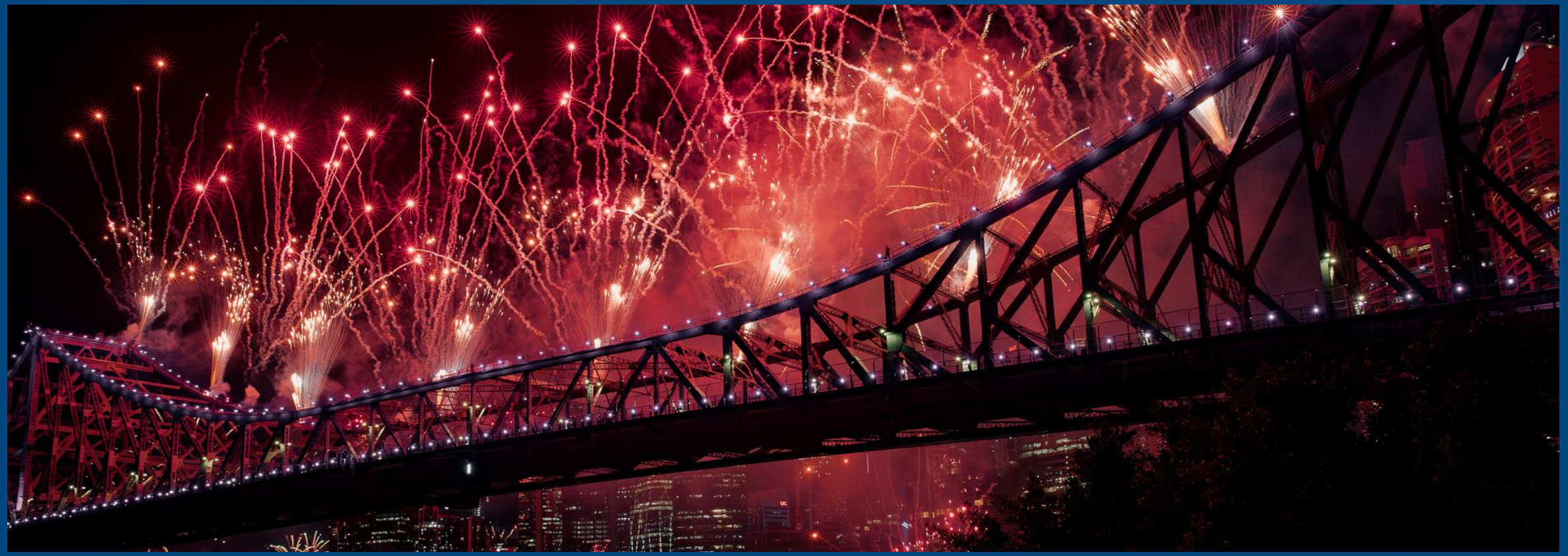
Dr. Krishna Manaswi Digumarti

School of Electrical Engineering and Robotics

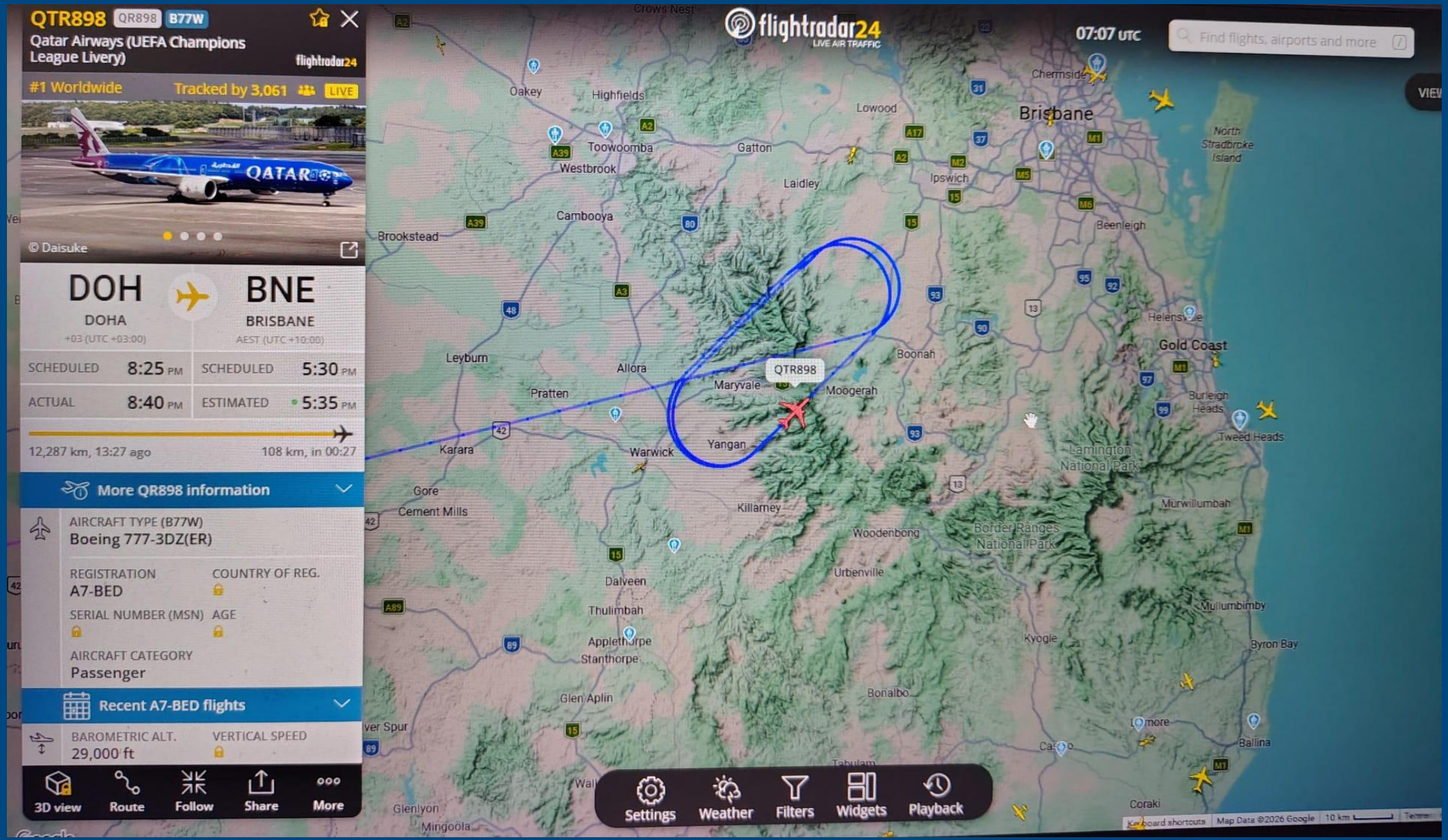
ACKNOWLEDGEMENT OF TRADITIONAL OWNERS

QUT acknowledges the Turrbal and Yugara, as the First Nations owners of the lands where QUT now stands. We pay respect to their Elders, lores, customs and creation spirits. We recognise that these lands have always been places of teaching, research and learning.

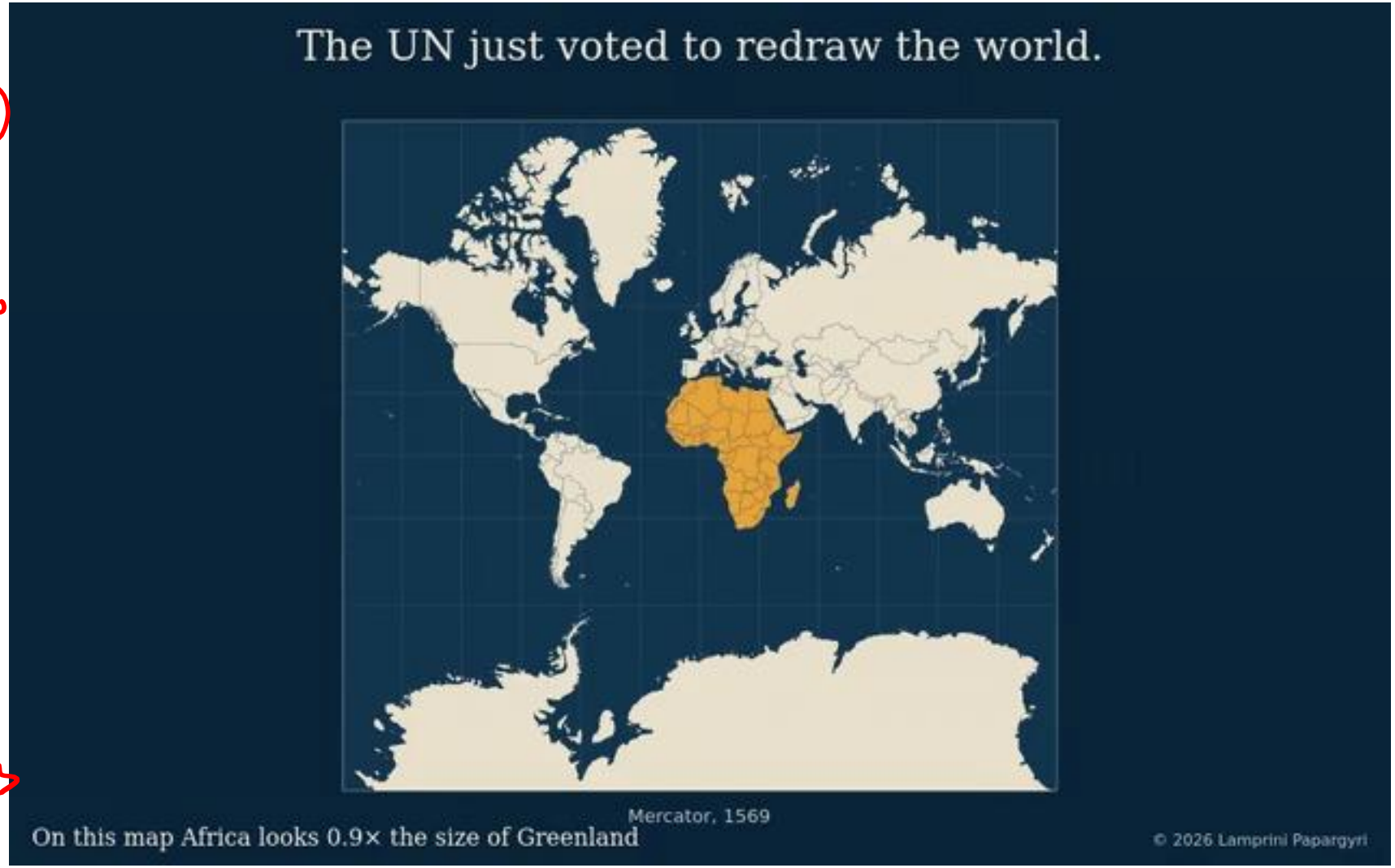
QUT acknowledges the important role Aboriginal and Torres Strait Islander people play within the QUT community.



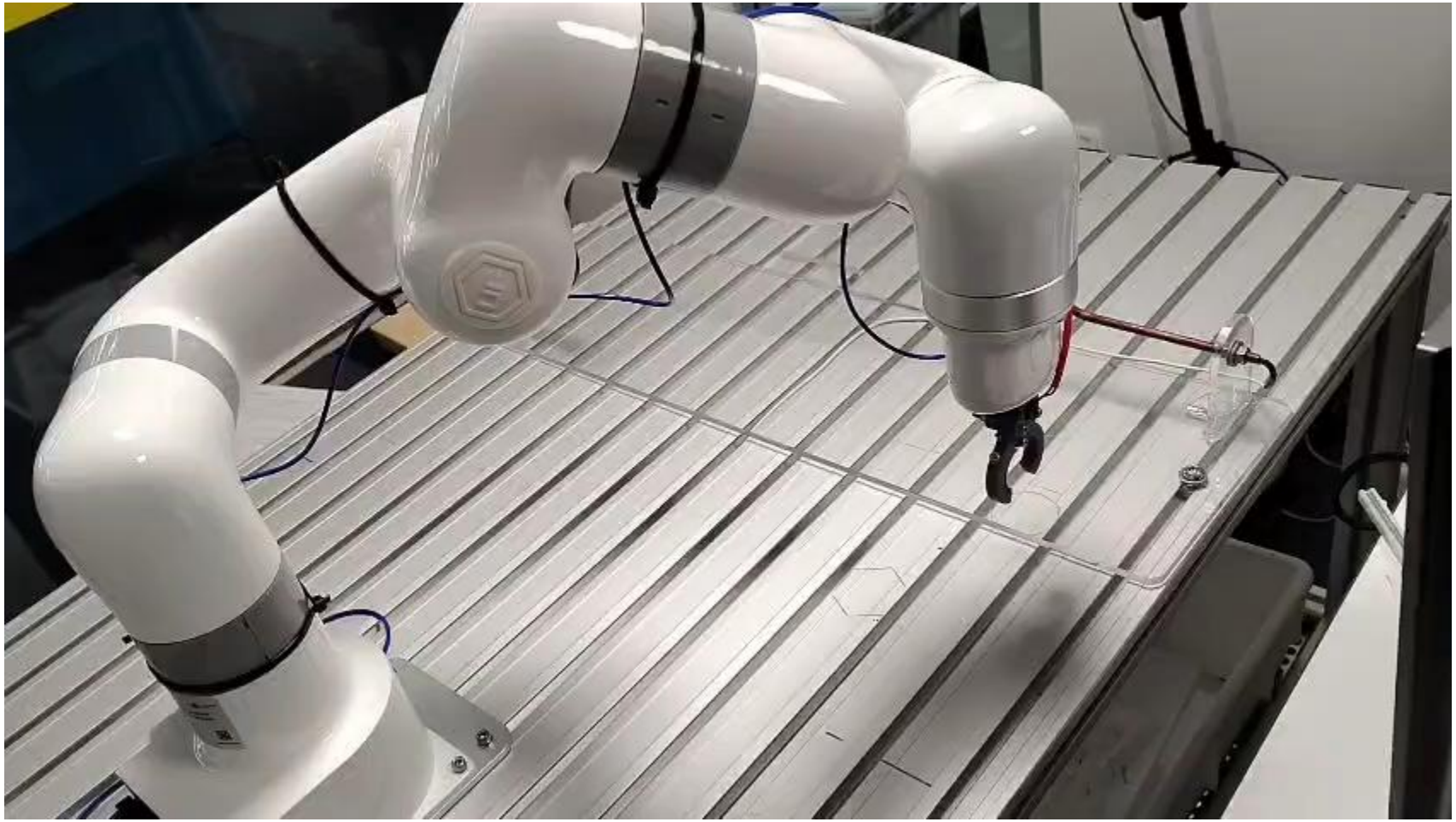
Credit: Howard Smith Warves



ENN581 Robot Motion, Control and Planning
Dr. KM Digumarti


The UN just voted to redraw the world.
Mercator, 1569
On this map Africa looks 0.9× the size of Greenland
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Assessment 2.3

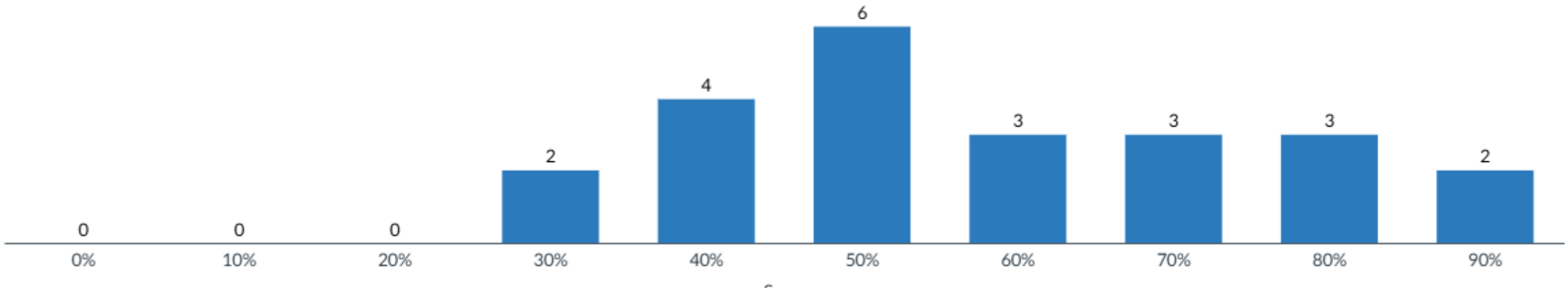


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Mid-semester exam

95%	35%	60%	55%	01:01:41
High Score	Low Score	Mean Score	Median Score	Mean Elapsed Time

Score Distribution



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What did we learn so far?

Forward Kinematics: Given the positions of the joints, where is the end effector?

$$^{0}\mathbf{T}_{n} \text{ or } \boldsymbol{\xi}_{n} = \mathcal{K}(\mathbf{q}),\ \mathbf{q} = \{\mathrm{q}_{i}, i \in [1, \cdots, n]\}$$

Inverse Kinematics: Given the tool pose, what are the joint positions to get there?

$$\mathbf{q} = \mathcal{K}^{-1}(\boldsymbol{\xi}_{n})$$

Velocity Kinematics: How fast is the robot moving (FK) or should it go (IK)?

$$^{0}\boldsymbol{\nu} = {}^{0}\mathbf{J}(\mathbf{q})\dot{\mathbf{q}}$$

Trajectory generation: Along what path should a manipulator move?

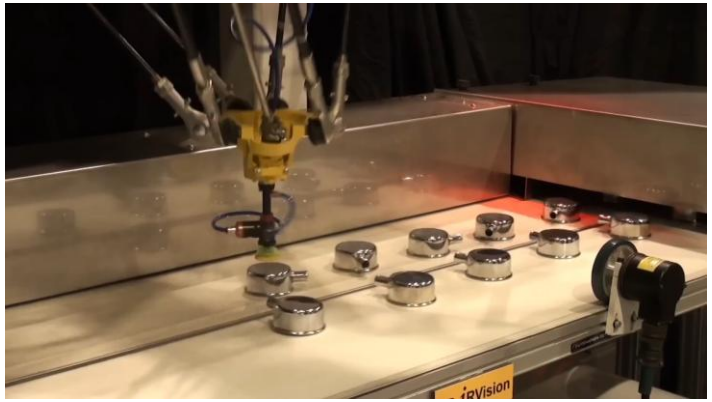
$$\mathbf{q}(t) \text{ and } \boldsymbol{\xi}_{\boldsymbol{n}}(t)$$

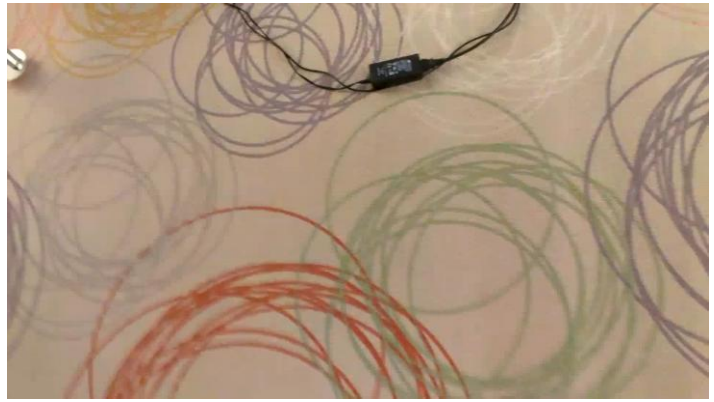
Dynamics (inverse): What are the torques at the joint, given a trajectory?

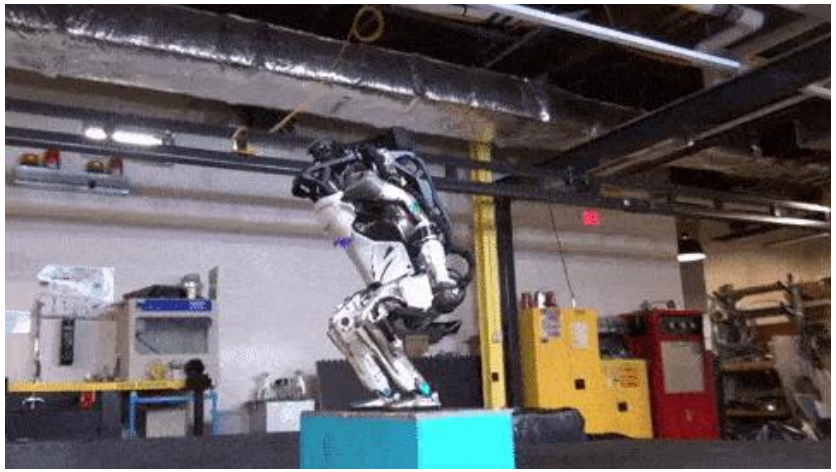
$$\mathbf{Q} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \boldsymbol{g}(\mathbf{q})$$

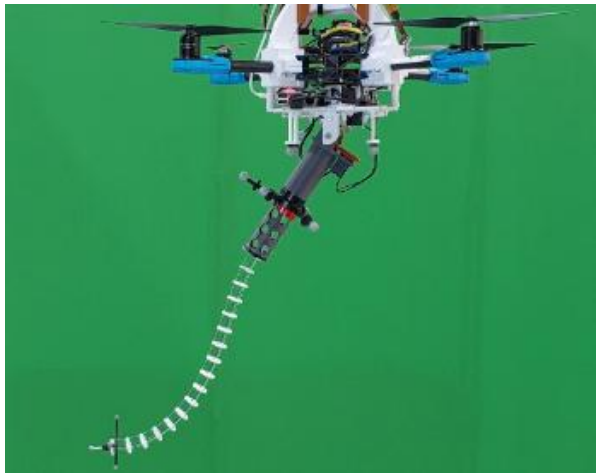
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ABB
Fanta









Today's lecture

Given a trajectory, we know how to compute required joint torques.

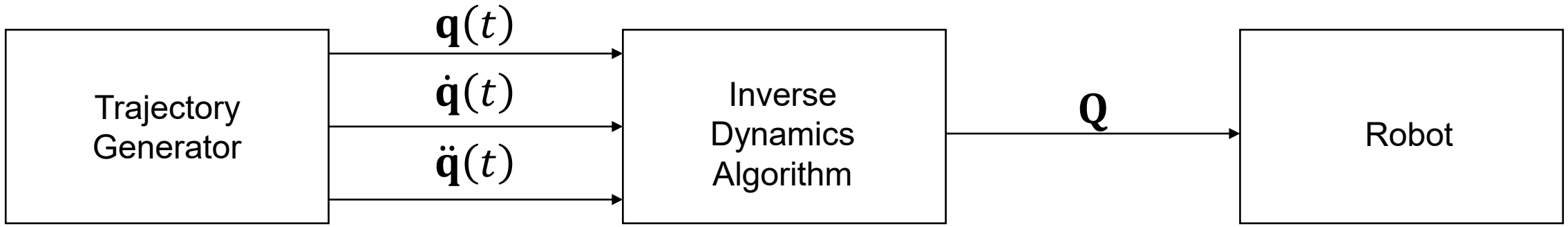
How do we ensure that the robot's joints move with the desired torques?

Today, we focus on trajectory following using concepts of linear control.

We will deal with end effector forces in the next lecture.

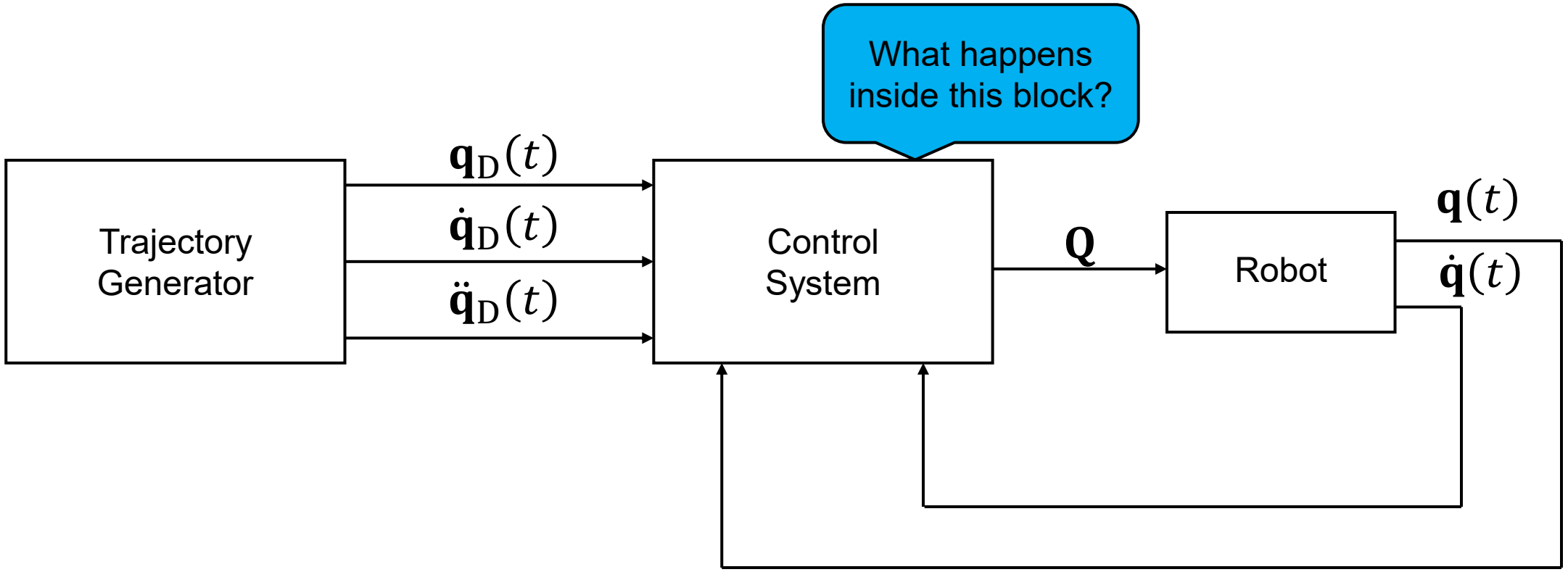
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Open-loop control



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Closed-loop control



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Servo error

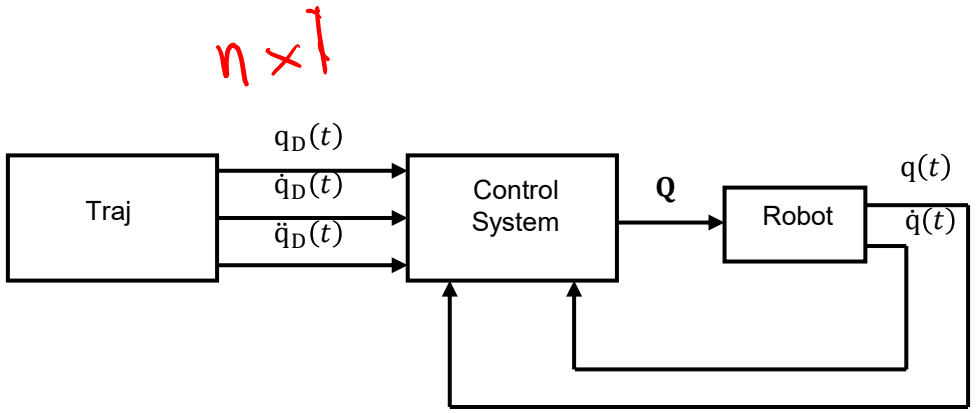
The error between the desired and actual positions and velocities

$$E = \mathbf{q}_{\mathrm{d}} - \mathbf{q}$$

The goal of designing a control system is to ensure that this error quickly tends to zero (or a small value) as time passes.

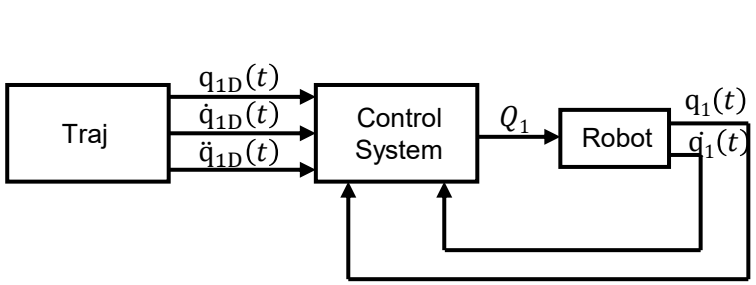
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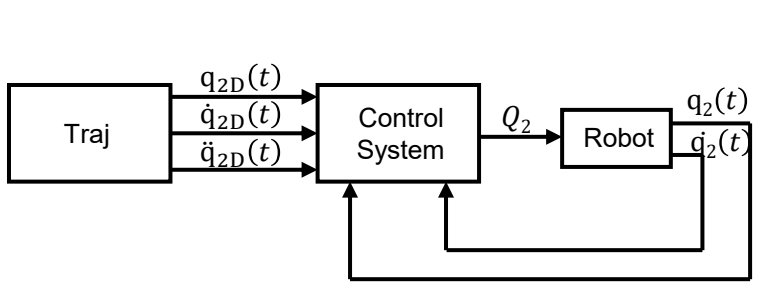
Independent joint control

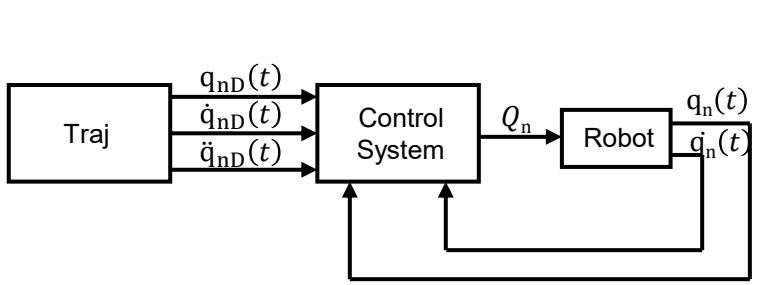


In general, for a robot with n joints, each line represents an $n \times 1$ vector.

We simplify this into n independent control problems where each line is a single entity.





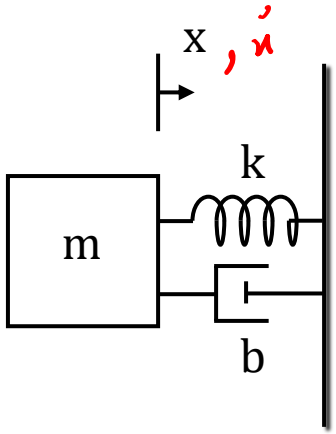


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A brief recap of control systems

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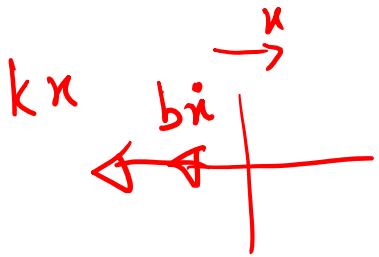
Consider this example of a spring-mass-damper system



The equation of motion of this system can be written as

$$m\ddot{x} + b\dot{x} + kx = 0$$

This is a second-order, homogeneous, linear differential equation with constant coefficients.



$m\ddot{x} = -kx - b\dot{x}$

The solution of this equation is the position of the block as a function of time, $x(t)$

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Let us solve the differential equation

$x(t) :=$

$$m\ddot{x} + b\dot{x} + kx = 0$$

Its characteristic equation is

$$ms^2 + bs + k = 0$$

The two roots of the equation (poles) are given as

$$s_1 = \frac{-b + \sqrt{b^2 - 4mk}}{2m} \qquad s_2 = \frac{-b - \sqrt{b^2 - 4mk}}{2m}$$



Pierre-Simon Laplace

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Case 1: roots are real and unequal

In this case, the solution is of the form

$$\mathrm{x}(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

where, A_1 and A_2 depend on initial conditions for position and velocity

and s_1 and s_2 are the roots of the characteristic equation

$$s_1 = \frac{-\mathrm{b} + \sqrt{\mathrm{b}^2 - 4\mathrm{mk}}}{2\mathrm{m}} \qquad s_2 = \frac{-\mathrm{b} - \sqrt{\mathrm{b}^2 - 4\mathrm{mk}}}{2\mathrm{m}}$$

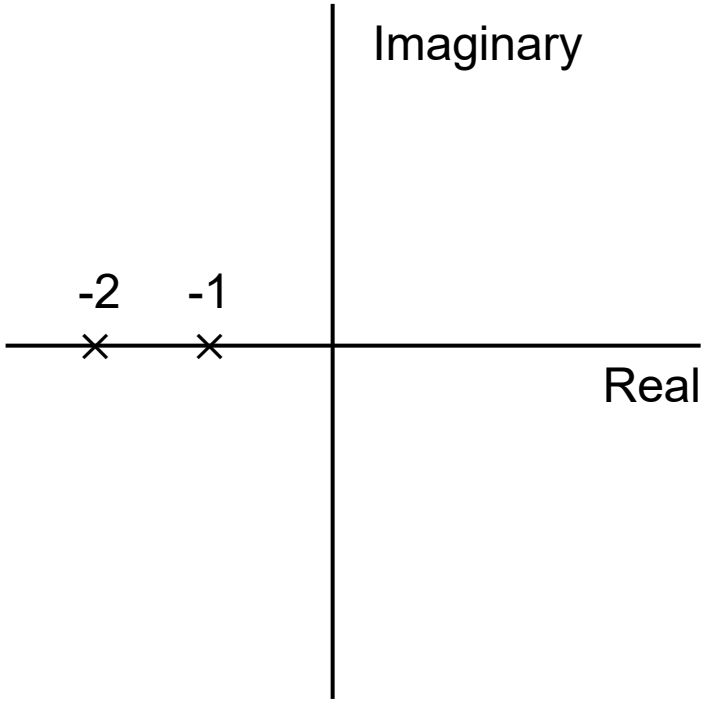
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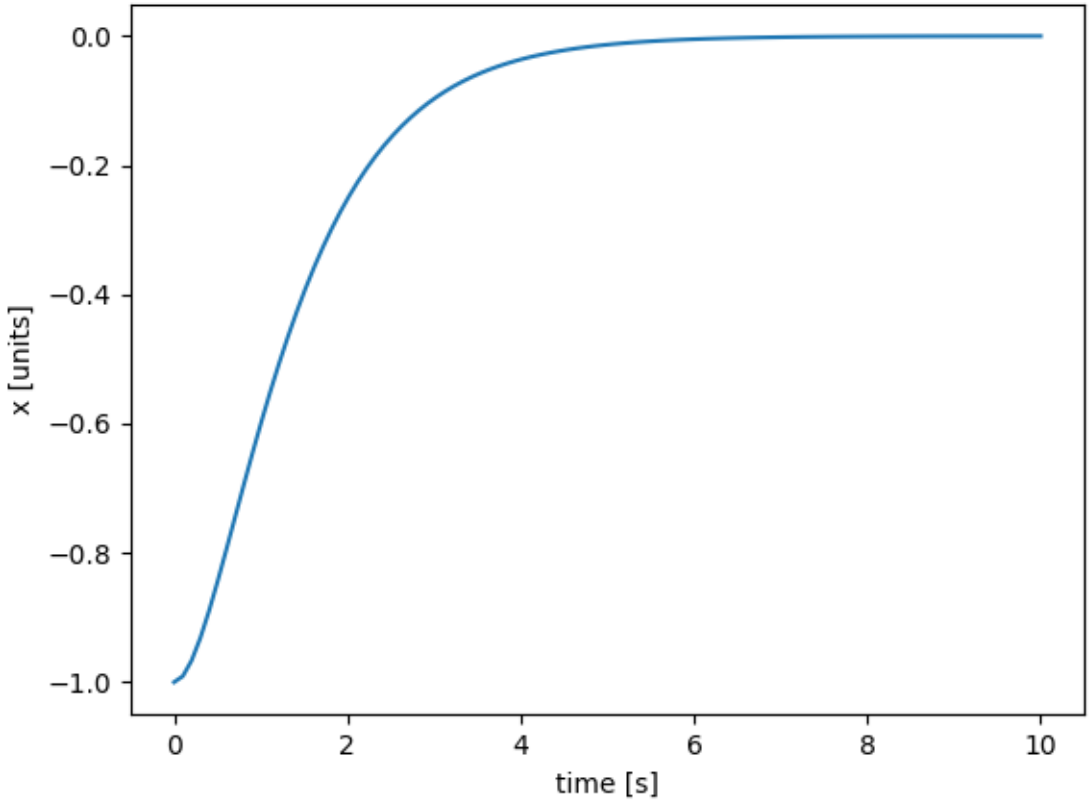
Example: real and unequal roots

- The parameters of the system are given as $\mathrm{m} = 1$, $\mathrm{b} = 3$ and $\mathrm{k} = 2$.
- Initial conditions are given as $\mathrm{x}(0) = -1$ and $\dot{\mathrm{x}}(0) = 0$.
- Determine the motion of the system.
- The roots are $s_1 = -1$ and $s_2 = -2$.
- The response of the system is $\mathrm{x}(t) = A_1 e^{-t} + A_2 e^{-2t}$.
- Using initial conditions gives $A_1 + A_2 = -1$ and $-A_1 - 2A_2 = 0$.
- Solving for these gives $A_1 = -2$ and $A_2 = 1$.
- The motion of the system for $t > 0$ is therefore $\mathrm{x}(t) = -2e^{-t} + e^{-2t}$.

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Plot the system response





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Case 2: Roots are complex

In this case, the solution is of the form

$$\mathrm{x}(t) = A_1 e^{\lambda t} \cos(\mu t) + A_2 e^{\lambda t} \sin(\mu t)$$

where, A_1 and A_2 depend on initial conditions for position and velocity

and s_1 and s_2 are the roots of the characteristic equation

$$s_1 = \lambda + \mathrm{i}\mu \qquad\qquad s_2 = \lambda - \mathrm{i}\mu$$

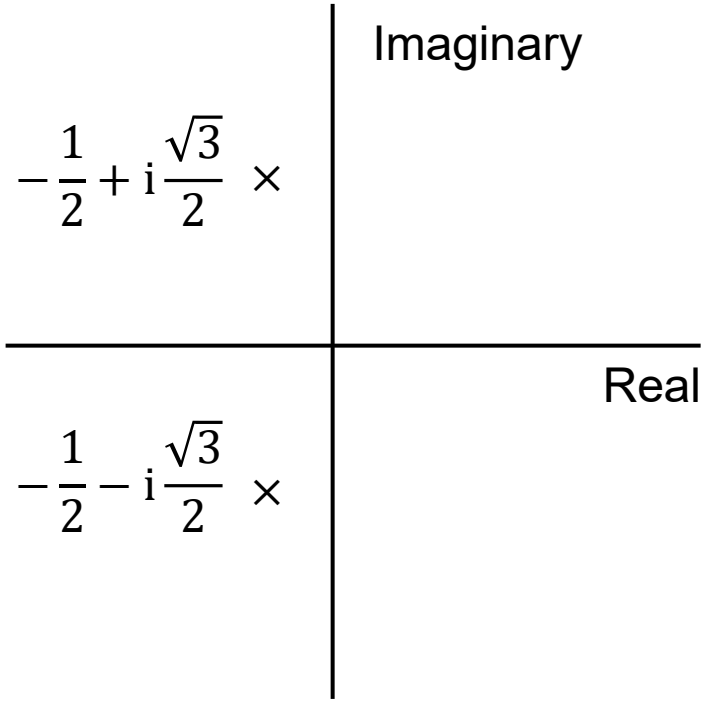
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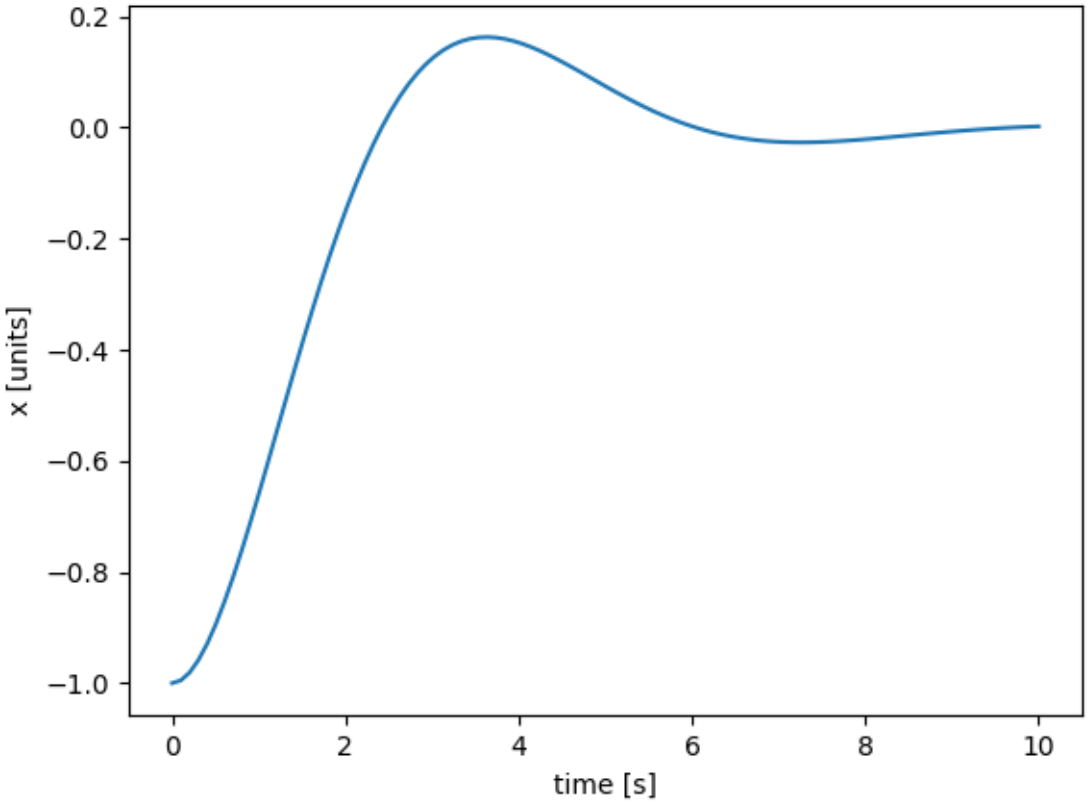
Example: complex roots

- The parameters of the system are given as $\mathrm{m} = 1$, $\mathrm{b} = 1$ and $\mathrm{k} = 1$.
- Initial conditions are given as $\mathrm{x}(0) = -1$ and $\dot{\mathrm{x}}(0) = 0$.
- Determine the motion of the system.
- The roots are $s_1 = -\frac{1}{2} + \mathrm{i}\frac{\sqrt{3}}{2}$ and $s_2 = -\frac{1}{2} - \mathrm{i}\frac{\sqrt{3}}{2}$.
- The response of the system is $\mathrm{x}(t) = A_1 e^{-t/2}\cos\left(\sqrt{3}t/2\right) + A_2 e^{-t/2}\sin\left(\sqrt{3}t/2\right)$.
- Using initial conditions gives $A_1 = -1$ and $\frac{-A_1}{2} + \sqrt{3}A_2/2 = 0$.
- Solving for these gives $A_1 = -1$ and $A_2 = -1/\sqrt{3}$.
- The motion of the system for $t > 0$ is therefore $\mathrm{x}(t) = -e^{-t/2}\left[\cos\left(\frac{\sqrt{3}t}{2}\right) + \left(\frac{1}{\sqrt{3}}\right)\sin\left(\frac{\sqrt{3}t}{2}\right)\right]$.

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Plot the system response





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Case 3: Roots are real and equal

In this case, the solution is of the form

$$\mathrm{x}(t) = (A_1 + A_2 t)e^{-\frac{\mathrm{b}}{2\mathrm{m}}t}$$

where, A_1 and A_2 depend on initial conditions for position and velocity

and s_1 and s_2 are the roots of the characteristic equation

$$s_1 = s_2 = -\frac{\mathrm{b}}{2\mathrm{m}}$$

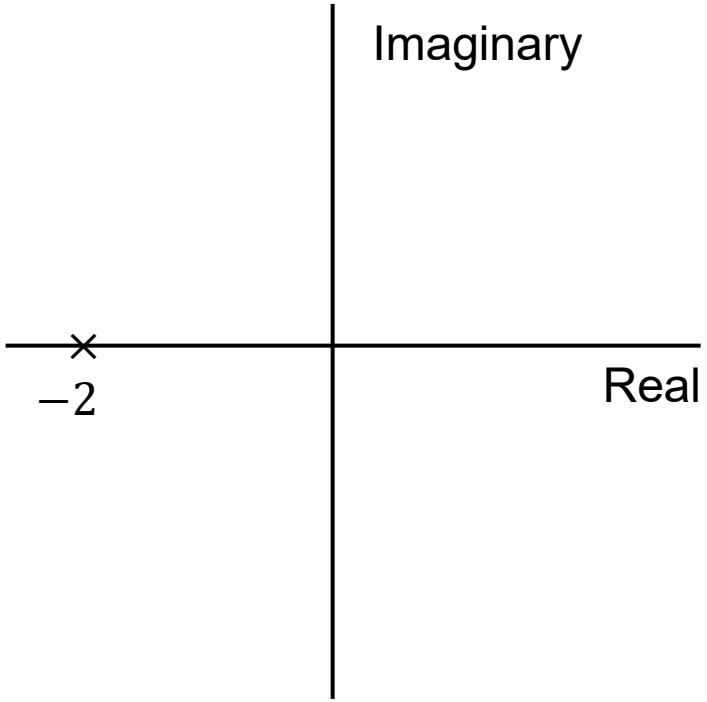
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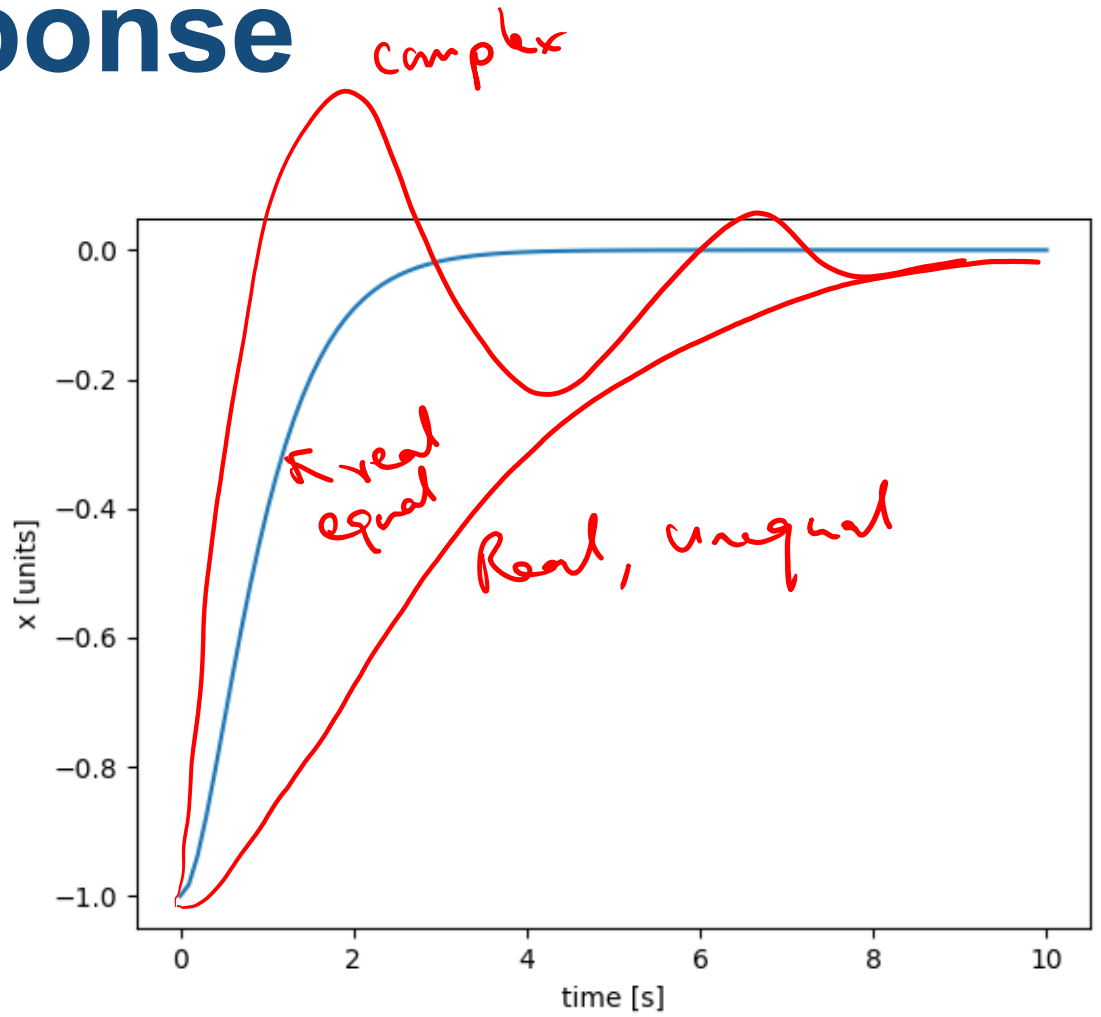
Example: real and equal roots

- The parameters of the system are given as $\mathrm{m} = 1$, $\mathrm{b} = 4$ and $\mathrm{k} = 4$.
- Initial conditions are given as $\mathrm{x}(0) = -1$ and $\dot{\mathrm{x}}(0) = 0$.
- Determine the motion of the system.
- The roots are $s_1 = -2$ and $s_2 = -2$.
- The response of the system is $\mathrm{x}(t) = (A_1 + A_2 t)e^{-2t}$.
- Using initial conditions gives $A_1 = -1$ and $-2A_1 + A_2 = 0$.
- Solving for these gives $A_1 = -1$ and $A_2 = -2$.
- The motion of the system for $t > 0$ is therefore $\mathrm{x}(t) = (-1 - 2t)e^{-2t}$.

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Plot the system response

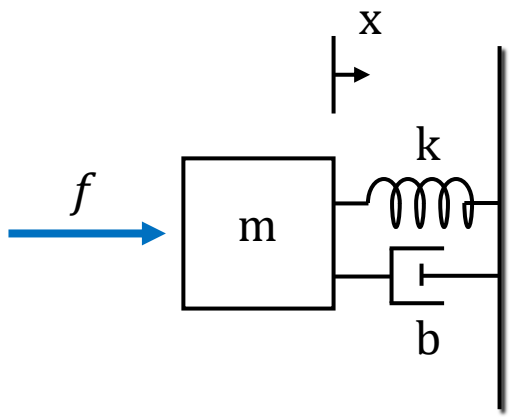




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How do we control such a system?

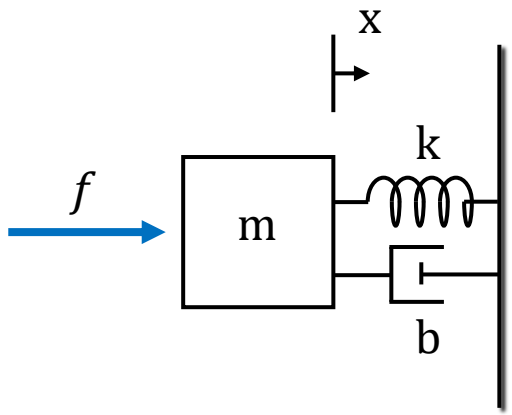
- So far, we looked at the open-loop dynamics of the system.
- Given: a spring-mass-damper system and an actuator.
- Aim: to make the system behave in a certain way (critically damped, for example). What force should the actuator apply?



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Equation of motion, with the actuator force is

$m\ddot{x} + b\dot{x} + kx = f$



$$m\ddot{x} + b\dot{x} + kx = f$$

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We would like to propose a formula for f

Assume that we incorporated sensors to obtain information of the position and velocity of the mass,

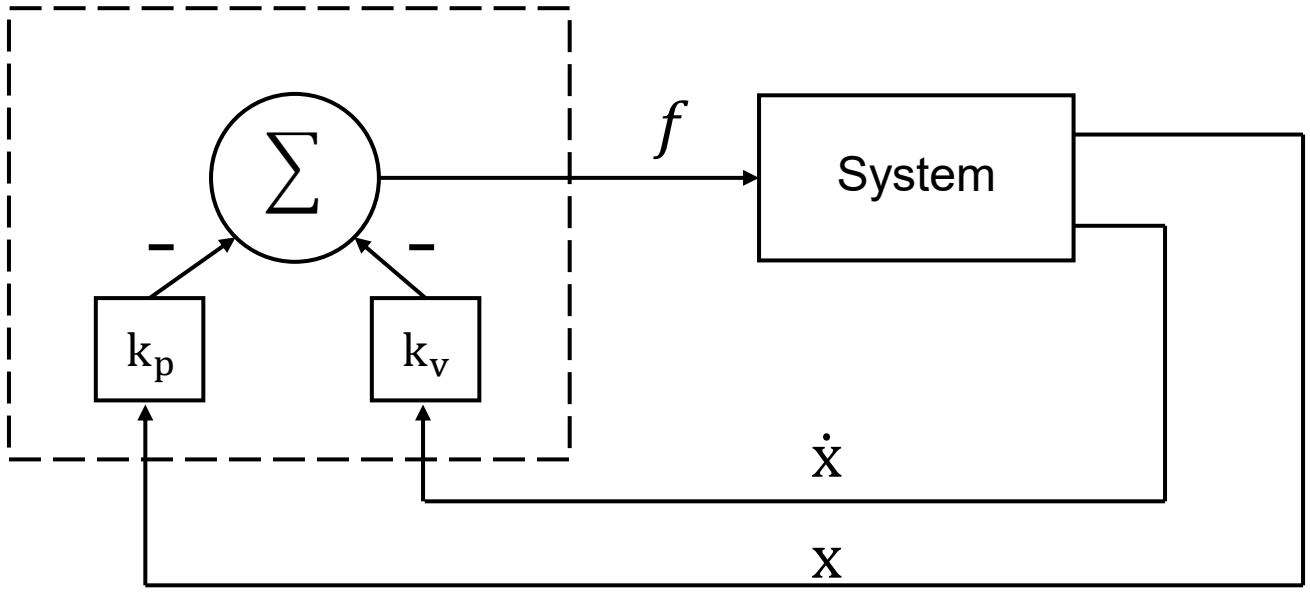
i.e., we measure $\mathrm{x}(t)$ and $\dot{\mathrm{x}}(t)$.

We can devise a formula for the actuator force that makes the mass move as we desire. An example of this formula, or control law, is given as follows.

$$f = -\mathrm{k_p x} - \mathrm{k_v}\dot{\mathrm{x}}$$

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Block diagram of the simple control law



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Choice of control gains determines system behaviour

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Choice of control gains determines system behaviour

$$m\ddot{x} + b\dot{x} + kx = f$$

$$m\ddot{x} + b\dot{x} + kx = -k_p x - k_v \dot{x}$$

$$m\ddot{x} + (b + k_v)\,\dot{x} + (k + k_p)x = 0$$

$$m\ddot{x} + \acute{b}\dot{x} + \acute{k}x = 0$$

By choosing the control gains k_p and k_v, we can tune the behaviour of the second order system.

Example: if k_p and k_v are chosen such that $\acute{b}^2 = 4m\acute{k}$, the system is critically damped.

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Control-law partitioning

- Control-law is simply the formula that we set for f
- It is common to partition this into two parts
  - one that depends on the model of the system (m, b and k) – model-based
  - the other that is independent of these – servo portion

$$f = \alpha \acute{f} + \beta$$

- α and β are functions or constants. They are chosen in such a way that if $\acute{f}$ is the new input to the system, the system appears to be of unit mass.

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Choosing α and β

$$m\ddot{x} + b\dot{x} + kx = \alpha\acute{f} + \beta$$

We choose $\alpha = m$ and $\beta = b\dot{x} + kx$, so that the above equation becomes

$$\ddot{x} = \acute{f}$$

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We can define a new control law for $\acute{f}$

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We can define a new control law for $\acute{f}$

$$\acute{f} = -k_v\dot{x} - k_p x$$

Now the equation becomes

$$\ddot{x} = \acute{f}$$

$$\ddot{x} = -k_v\dot{x} - k_p x$$

$$\ddot{x} + k_v\dot{x} + k_p x = 0$$

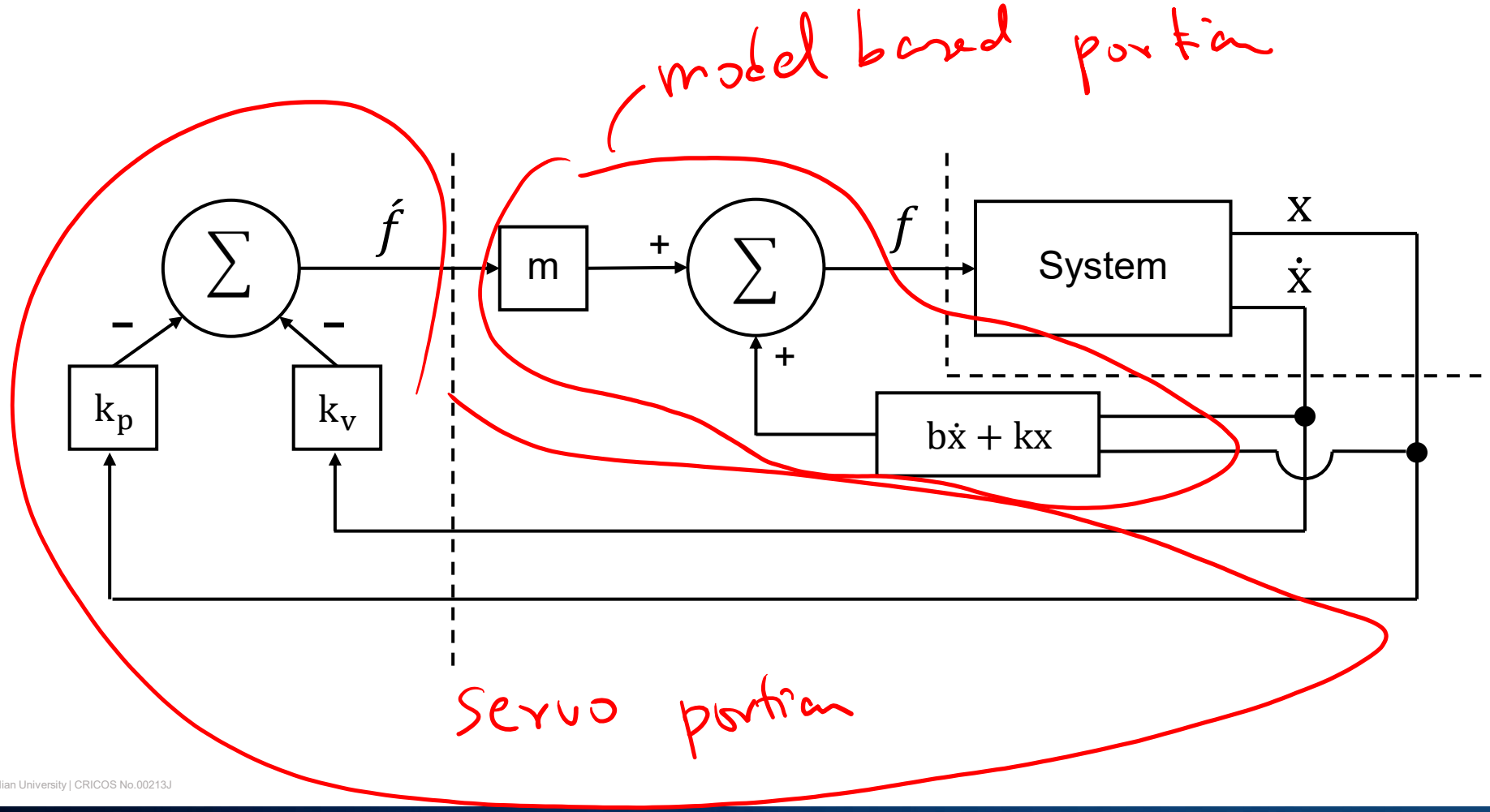
$$s^2 + k_v s + k_p = 0$$

These two terms are independent of m, b and k

For a critically damped system, we can directly write $k_p{}^2 = 4k_v$.

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Block diagram of the partitioned controller



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Trajectory following

- The control system that we designed just now, maintains the block at a fixed position x.
- Assume that we want to get the block to move in along a desired trajectory $\mathrm{x}_d(t)$.
- Assume that the trajectory is smooth and twice differentiable. This means that we can compute $\dot{\mathrm{x}}_d(t)$ and $\ddot{\mathrm{x}}_d(t)$.

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Define a new control law that incorporates servo error

Recall servo error: $E = x_d - x$

Control law that incorporates the servo error is

$$\acute{f} = \ddot{x}_d + k_v \dot{E} + k_p E$$

This results in the equation

$$\ddot{x} = \ddot{x}_d + k_v \dot{E} + k_p E$$

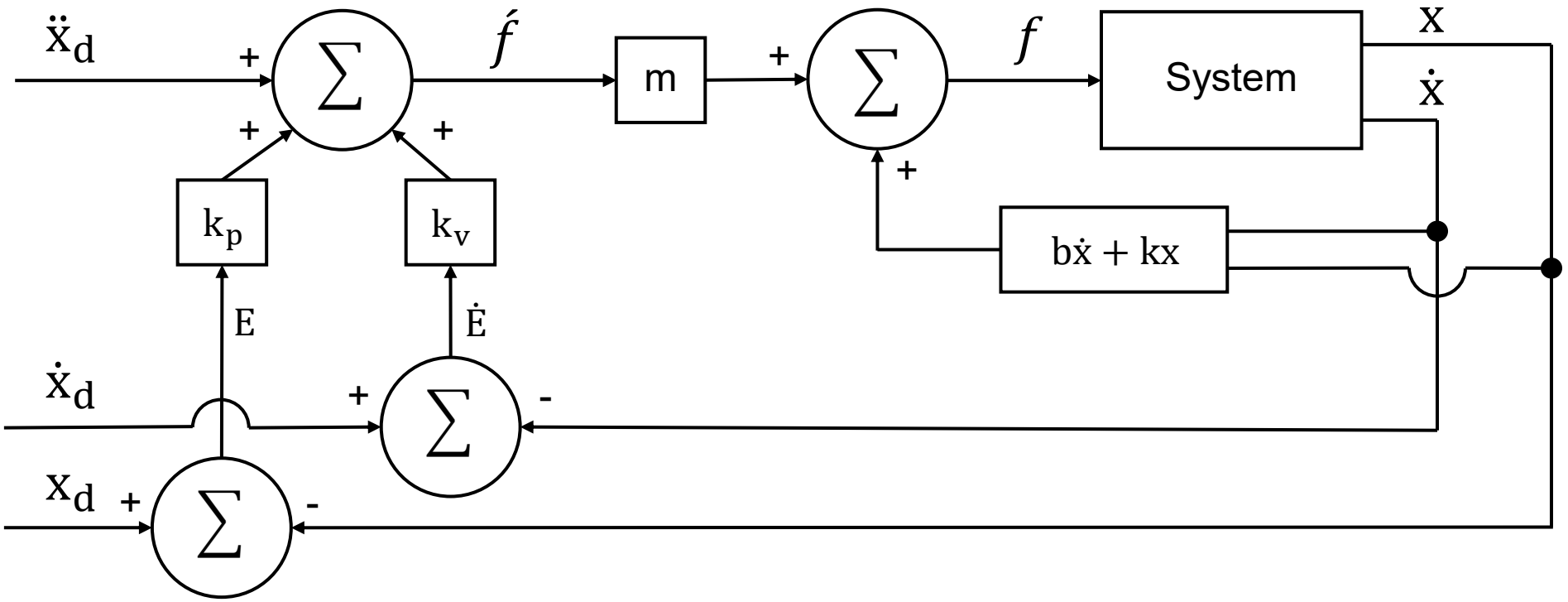
$$0 = \ddot{E} + k_v \dot{E} + k_p E$$

$\ddot{x}_d - \ddot{x}$

Second order in error space

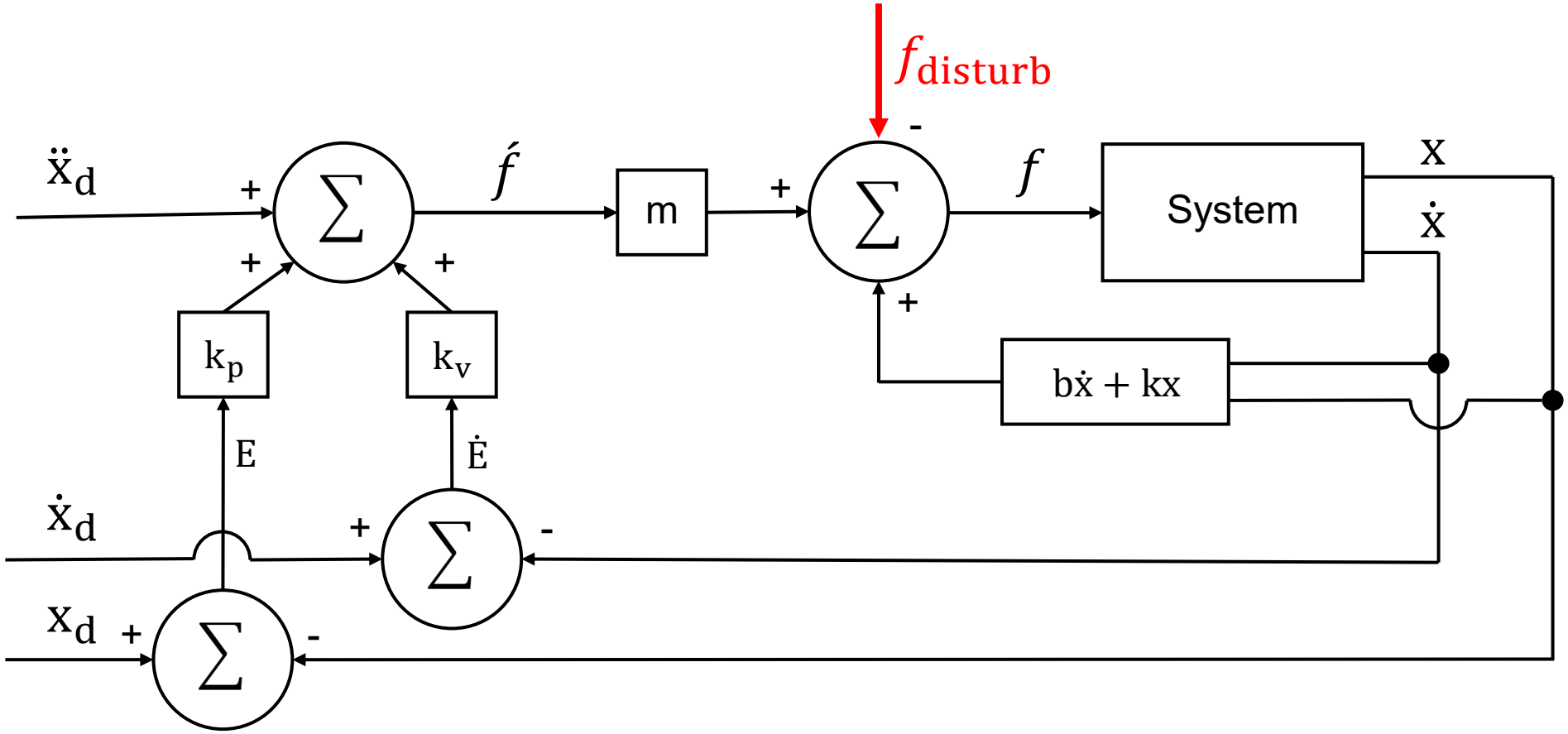
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Block diagram of the partitioned controller that incorporates servo error



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Disturbance rejection



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Error equation in the presence of disturbance

$$\ddot{\mathrm{E}} + \mathrm{k_v}\dot{\mathrm{E}} + \mathrm{k_p}\mathrm{E} = f_{\mathrm{disturb}}/\mathrm{m}$$

The bounded-input bounded-output principle states that if f_{disturb} is bounded, then the error E is also bounded.

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Steady state error

- Consider the special case when f_{disturb} is a constant value.
- We can analyse the system in it steady state i.e., when derivatives of the state are 0.

$$\ddot{E} + k_v\dot{E} + k_pE = f_{\text{disturb}}/m$$

(0 over $\ddot{E}$, 0 over $k_v\dot{E}$)

$$k_pE = f_{\text{disturb}}/m$$

Steady state error $\longrightarrow$ $$E = f_{\text{disturb}}/k_pm$$

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How do we get rid of steady state error?

Modify the control law to have an integral term

$$\acute{f} = \ddot{\mathrm{x}}_\mathrm{d} + \mathrm{k_v}\dot{\mathrm{E}} + \mathrm{k_p}\mathrm{E} + \mathrm{k_i}\int \mathrm{E}dt$$ ← PID control law

The system equation is now

$$\ddot{\mathrm{E}} + \mathrm{k_v}\dot{\mathrm{E}} + \mathrm{k_p}\mathrm{E} + \mathrm{k_i}\int \mathrm{E}dt = f_{\mathrm{disturb}}/\mathrm{m}$$

at steady state

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The system is now of third order

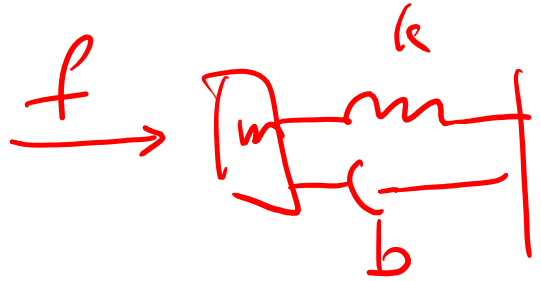
- If $\mathrm{E}(t) = 0$ for $t < 0$, then for $t > 0$, the system equation can be written as

$$\dddot{\mathrm{E}} + \mathrm{k_v}\ddot{\mathrm{E}} + \mathrm{k_p}\dot{\mathrm{E}} + \mathrm{k_i}\mathrm{E} = \dot{f}_{\mathrm{disturb}}/\mathrm{m}$$

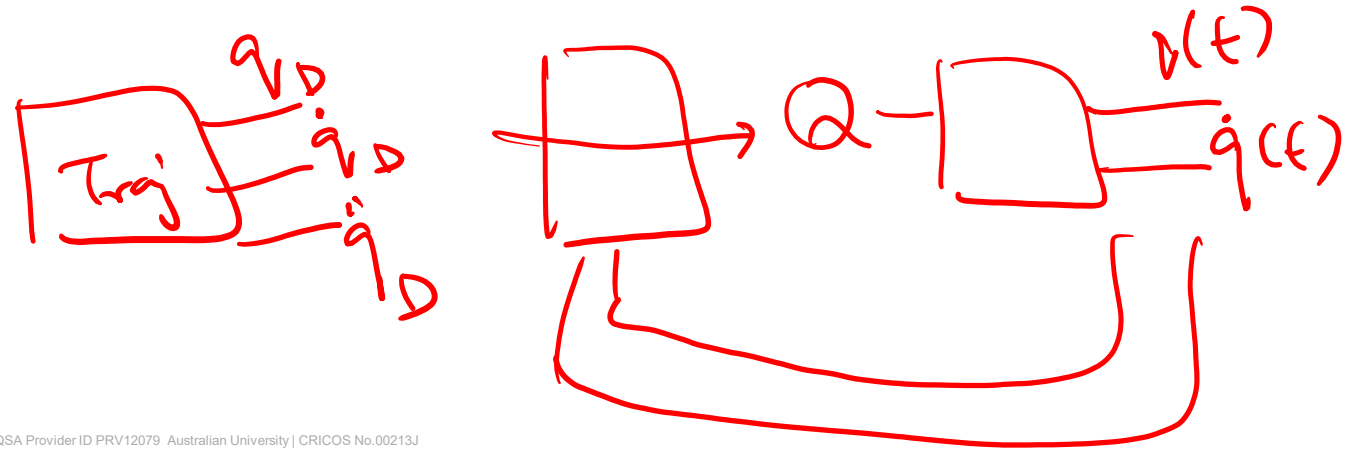
You can verify that at steady state, we obtain $\mathrm{E} = 0$

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Controlling a single joint



- A joint consists of an actuator connected to a link.
- In the previous lecture on dynamics, we determined the torque that is required at a joint for the link to have a desired position, velocity and acceleration profile.
- This torque is provided by an actuator – like a DC motor



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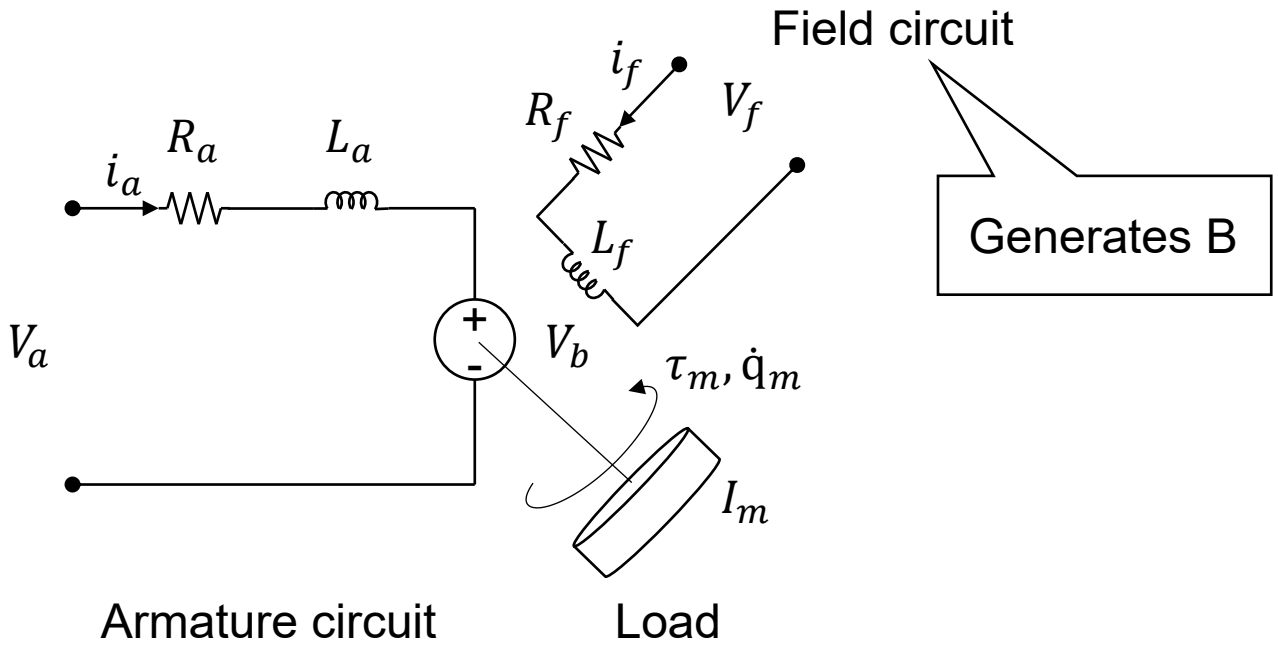
Modelling a joint

Recall our discussion of a DC motor from last semester

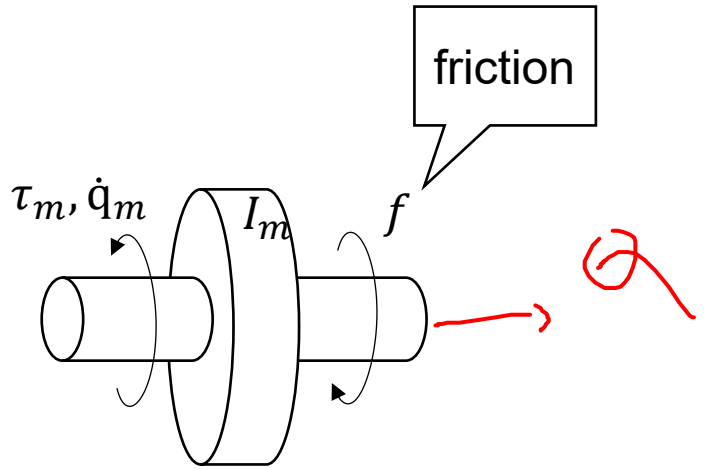
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A motor is an electromechanical device

Electrical model



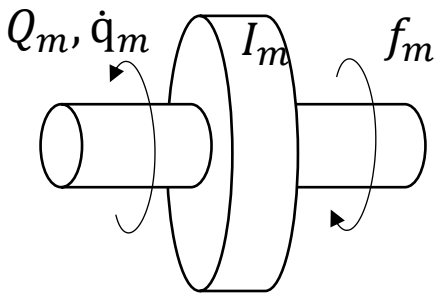
Mechanical model



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Current in armature coil generates a torque

$$Q_m(t) = k_1 B i_a = k_4 i_a$$



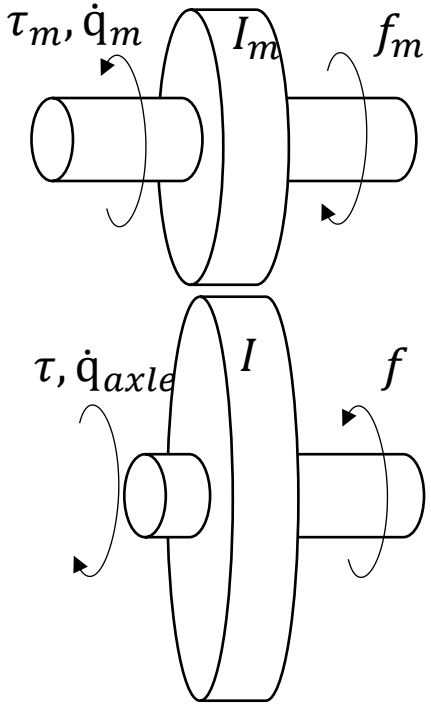
Consider there to be loss in output torque due to friction

$$Q_{net} = Q_m - f_m$$

$$I_m \ddot{q}_m = k_4 i_a - c_m \dot{q}_m$$

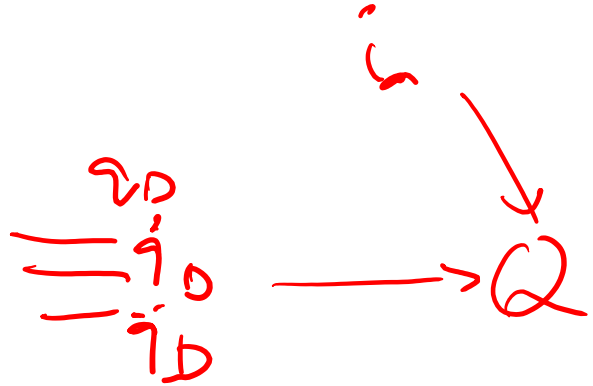
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What if gearing is included at the joint?



Gear ratio

$$Q = \eta Q_m$$



$$\dot{q}_{axle} = \frac{1}{\eta}\dot{q}_m$$

$$Q = \left(I + \eta^2 I_m\right)\ddot{q}_{axle} + \left(c + \eta^2 c_m\right)\dot{q}_{axle}$$

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Bringing it all together

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Summary

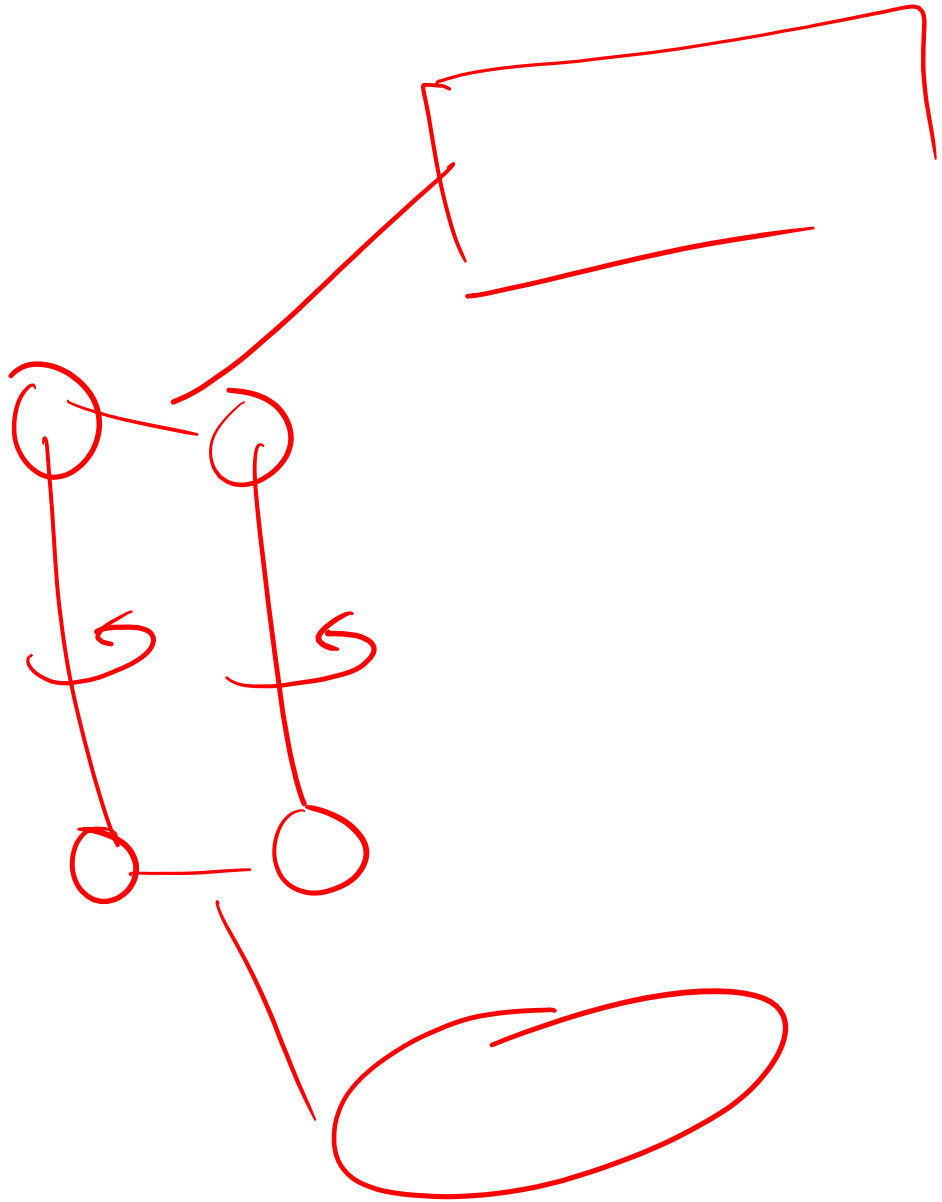
- A quick review of control systems
- Over damped, under damped and critically damped systems
- PID Control for trajectory following
- Modelling a joint using the electromechanical model of a DC motor

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Next week

Non linear models for control

Force control



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